Model answers

Question 1

1.a.i)

	Capacity factor
Wind site A	0.35
Wind site B	0.35
Wind site C	0.34

[2]

1.a.ii)

Overestimates because of the assumption that the wind farm and/or individual wind turbines have no down time for maintenance and/or force outages.
We are also unclear how the leap year is included in the 5-year measurements. If the first and last years are leap years, leap years are overrepresented in the averages as they are included in 2 out of 5 years. On the other hand, if the first year is not a leap year, leap years are underrepresented in the averages as they are included in 1 out of the 5 years. In practice, the leap year should be included 1 out of 4 years.
There are other assumptions about the type of wind turbines etc. which could go either way.
If a good argument is made, either answer is valid as is don't know and will be given full marks [2]

1.a.iii)

	Energy per year (TWh)
Wind site A	3.066
Wind site B	3.066
Wind site C	3.019

Note if a student uses the capacity factor for wind site C, the answer is 2.978 TWh.

[2]

1.a.iv)
The wind farm at site C will have highest LCOE because costs are the same, but energy will be lower. The wind farms at site A & B will have identical LCOEs. [2]

1.a.v)

The wind farm with the least benefit to the power system is Wind Site C because its LCOE is highest. [2]

1.a.vi)
The additional major assumption is that the cost of transmission to connect in each of these wind farms is also identical (or differences are small enough that they will now impact on the LCOE) and that there is no curtailment for any reason transmission and/or operational. The LCOE is also based on the capacity factors above so the over or underestimate of the capacity factor needs to be accounted for. The addition of transmission costs will make the LCOE higher in all cases and the capacity factor overestimate will lead to the LCOE being higher so for both assumptions they underestimate the LCOE. [5]

1.b.i)

Answer = sum of values in Table 2 *365=176 GWh*365 *(1/1000 TWh/GWh)= 64.24 TWh [2]

1.b.ii)

	Midnight to 6AM	6AM to Noon	Noon to 6PM	6PM to Midnight
Wind site A (MWh)	1980	2040	2100	2280
Wind site B (MWh)	2040	2100	2280	1980
Wind site C (MWh)	2100	2030	2280	1860
Demand (MWh)	8200*3	15000*3	22333.33*3	13133.33*3

[3]

The most beneficial hours are at the peak so all other things being equal the wind farm that has the highest output at the peak is likely to be the most beneficial. The peak is 2 to 4PM which is in the middle of the six-hour window of noon to 6PM. Wind site A is almost 10 % lower in output than site B & C during this period and so in this analysis (capacity value) A is least valuable and B & C are equally valuable BUT in addition if we consider the capacity factor then site C is not as valuable as B. Wind site B is the most beneficial overall. Students may plot out normalized profiles and eyeball them to see if there is more or less correlation – this is also a valid methodology.

[3]

1.b.iii)

Answer = 3.066/64.24 = 4.77 % [2]

1.c)

From midnight to 2AM the average demand is 8.4/2 GW = 4.2GW and with a base loaded nuclear at 4GW then on average pr hour there is 0.2GW to be supplied by wind, wind at site B can produce 2040/6 MW (0.34 GW). 140 MW per hour is curtailed. Hence 280 MWh curtailed for 2 hours. [2]

From 4AM to 6AM the average per hour demand is 8.2/2 GW = 4.1GW and with a base loaded nuclear at 4GW then on average per hour there is 0.1GW to be supplied by wind. Wind can produce 0.34 GW. Hence, 240 MW per hour is curtailed. 480 MWh. [2]

From 2 to 4 am, it is all curtailed, 340MW *2 hours=

Average wind curtailment per day =0.28+0.48+0.68=1.44 GWh [2]

Average wind curtailment per year = 1.44*365 = 525.6 GWh [2]

Question 2

2.a.i)

New decision variables:

RP: Rated electric power (in MW) – maximum electricity consumption of the electric equipment 1 mark

WC: Warehouse capacity (in units of good) 1 mark

Prod_t: Production of industrial plan during time period t (in units of good) 0.5 mark

Load_t: Number of goods loaded in the storage during time period t (in units of good) 0.5 mark

Unload_t: Number of goods unloaded from storage during time period t (in units of good) 0.5 mark

Balance_t: Number of goods stored in the warehouse at the end of time period t — 1 mark
Sales_t: Sale of goods during each time period t — 0.5 mark

2.a.ii)

New objective function terms
CAP_E · RP: Capital cost for industrial electrical equipment — 1 mark
CAP_W · WC: Capital cost for warehouse — 1 mark
FOM_I · WC: Fixed operating costs for temperature control in the warehouse — 1 mark
VC_L ·Load_t: Cost for loading storage — 1 mark
VC_U ·Unload_t: Cost for un-loading storage — 1 mark

2.a.iii)

New constraints:
Prod_t <= η· RP, production of goods cannot exceed the maximum production at rated capacity. — 1 mark
Balance_t<=WC, the number of goods stored cannot exceed the warehouse capacity — 1 mark
Balance_t= Balance_t-1+ Load_t- Unload_t, the number of goods in the warehouse changes based on loading and unloading schedule — 1 mark
Prod_t+ Unload_t= Load_t +Sales_t, The production plus unloading is equal to the sales and loading of the product — 1 mark
Sum_t (I_d,t · sales_t)>=Daily_demand_d, The sales over a day have to be higher than the fixed demand. — 1 mark

2.a.iv)

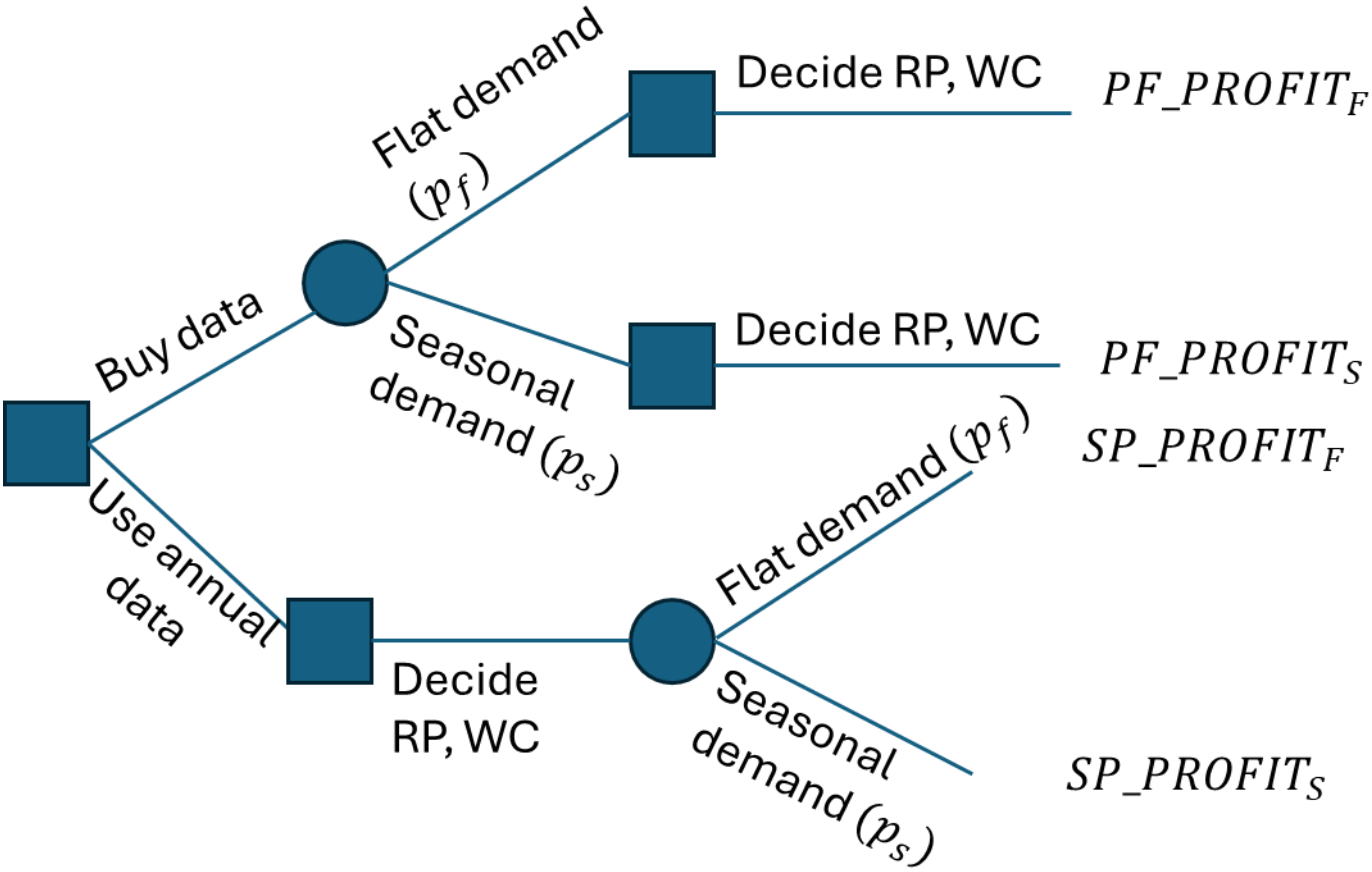
η: efficiency of equipment (number of goods/MW)
CAP_E: Capital cost for industrial electrical equipment (£/MW) — 1 mark
CAP_W: Capital cost for warehouse (£/maximum number of goods that can be stored) — 1 mark
FOM_I: Fixed operating costs for temperature control in the warehouse (£/maximum number of goods that can be stored) — 1 mark
VC_L: Cost for loading storage (£/number of goods) — 0.5 mark
VC_U: Cost for un-loading storage (£/number of goods) — 0.5 mark
Daily_demand_d: Daily demand (number of goods) — 0.5 mark
I_d,t: Indicator that has value 1 if time t belongs to day d — 0.5 mark

2.b.i)

First-stage decision variables are the decision variables that govern investment decisions (i.e., RP, WC) — 2 marks
Second-stage decision variables are the decision variables that can be adjusted depending on the system conditions. Here, the industry would aim to produce during the hours with lowest electricity price (i.e., all the variables indexed by t will also need to be indexed by s) — 3 marks

2.c.i)

Round used for chance nodes — 1 mark
Square used for decision nodes — 1 mark
Recognizing that decisions are made after uncertainty clears when buying the data — 2 marks



2.c.ii) The information will be more or equally valuable before investment decisions are made because there are more decisions that can potentially change.

2.c.iii) The capital and operating costs would be the same because in both cases it would choose the RP and production schedule that would minimize the electricity costs and manage fluctuations in demand through different inventory policies. 2 marks

The value of information will be the same as only operational decisions would change 1 mark

Question 3

3.a.i)

The expected value of the perfect foresight solutions will be (700+600+500)/3=600£.

2 marks

3.a.ii) Given that we do not have the best bound from the solver, we cannot calculate the MIP gap precisely. Nevertheless, the PF solutions give us a lower bound for the SP. Hence, we can use this to estimate the MIP gap (800-600)/800=25%

1 mark for correct formula for MIP gap

1 mark for correct calculation

3.a.iii)

VSS=EEV-SP*

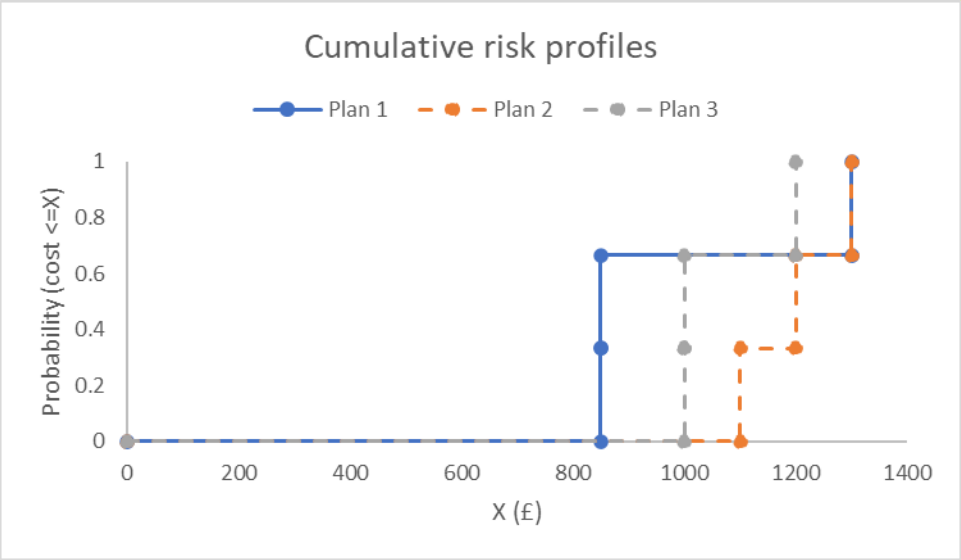
1 mark for correct formula
SP*<= 800£
2 marks
EEV-SP*>=100£.
1 mark for providing 100 as the value but without being clear if it is a lower bound.

3.a.iv)

EVPI=SP*-PF
1 mark for correct formula
SP*<=800£
EVPI<=200£
1 mark for providing 200 as the answer
2 marks for correctly identifying it as an upper bound

3.b.i)

X axis label and units 1
Y axis label and units 1
Graphs 1



3.b.ii) There is no deterministic dominance because the ranges of the outcomes (costs) are overlapping. 1 mark

Stochastic dominance has to be evaluated in pairs:
(Plan 1, Plan 2) -> Plan 1 stochastically dominates Plan 2 1 mark
(Plan 1, Plan 3) -> no stochastic dominance; profiles crossing 1 mark
(Plan 2, Plan 3) → Plan 3 stochastically dominates Plan 2 1 mark

3.b.iii)

Correct regret formula [1 mark]
Regret for plan 1 per scenario: (100,0,0) -> max regret 100 [1/3 mark]
Regret for plan 2 per scenario: (100,350,250) -> max regret 350 [1/3 mark]
Regret for plan 3 per scenario: (0,150,150) -> max regret 150 [1/3 mark]
Min max regret: plan 1 [1 mark]

3.b.iv)

1 mark for recognizing that the regret would be calculated with respect to those plans
Regret for plan 1 per scenario: (600,250,350)
Regret for plan 2 per scenario: (600, 600,600)
Regret for plan 3 per scenario: (500,400,500)

Yes, plan 3 would be selected. [1 mark]

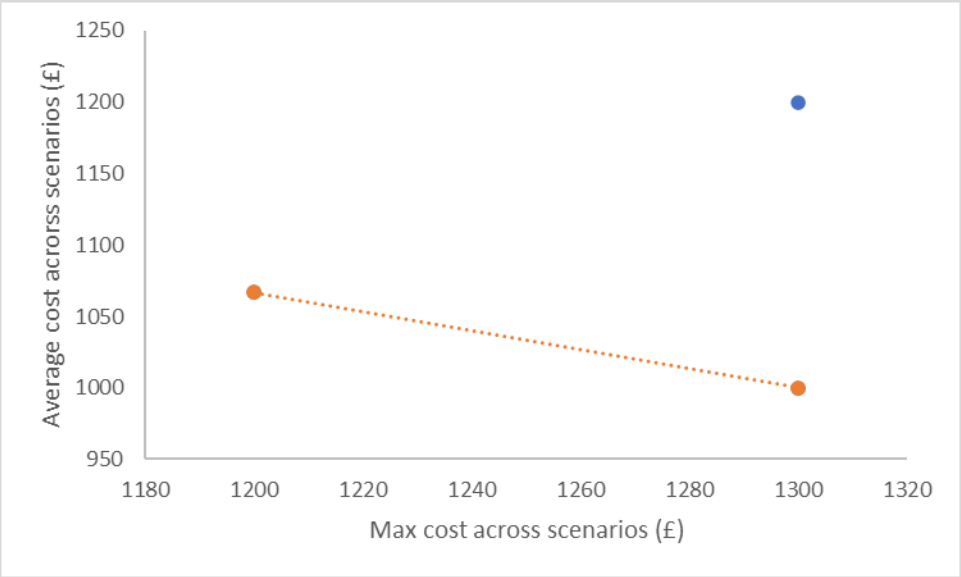
3.b.v)

No, because information from table 3 are irrelevant alternatives. 3 marks

3.b.vi)

Correct calculations for points 1 mark

Trade-off curve (orange line) 2 marks



3.b.vii)

I would present at least plan 1 and plan 3 because they are non-dominated solutions. It is beneficial to present more than 1 plans to a community because presentation of 1 plan leads to polarization between the two answers (yes/no). On the contrary, presenting multiple plans make clear trade-offs among plans and facilitate dialogue with a community.

3 marks